# Osaka Feedback Model II: Modeling Supernovae Based On High Resolution Simulations

Yuri Oku (Osaka U.)

Collaborators:
Kengo Tomida (Tohoku U.)
Kentaro Nagamine (Osaka U.)
Ikkoh Shimizu (Shikoku Gakuin U.)
Renyue Cen (Princeton U.)

# Introduction: Supernova Feedback

- Supernova feedback (SN FB) drive outflow and quench star formation.
- SN scale is unresolved in galaxy simulations.
- What do we need to model?

Credit: X-ray: NASA/CXC/JHU/D.Strickland; Optical: NASA/ESA/STScI/AURA/The Hubble Heritage Team; IR: NASA/JPL-Caltech/Univ. of AZ/C. Engelbracht

# SN Feedback Model: Kinetic & Thermal

- Two forms of feedback: **Kinetic** & **Thermal**

| | **Kinetic FB** | **Thermal FB** |
|---|---|---|
| Physical meaning | SNR gives momentum to ISM | Shock heating by SN ejecta |
| Roles | Drive **turbulence**,<br>**Expel gas** out from galaxy | Produce **hot outflow**,<br>**Transport metals** out from galaxy |
| Difficulty in low reso. | Unresolved cooling length,<br>Unresolved thin shell formation<br>(Due to low **spatial resolution**) | Unresolved cooling mass<br>(Due to low **mass resolution**) |

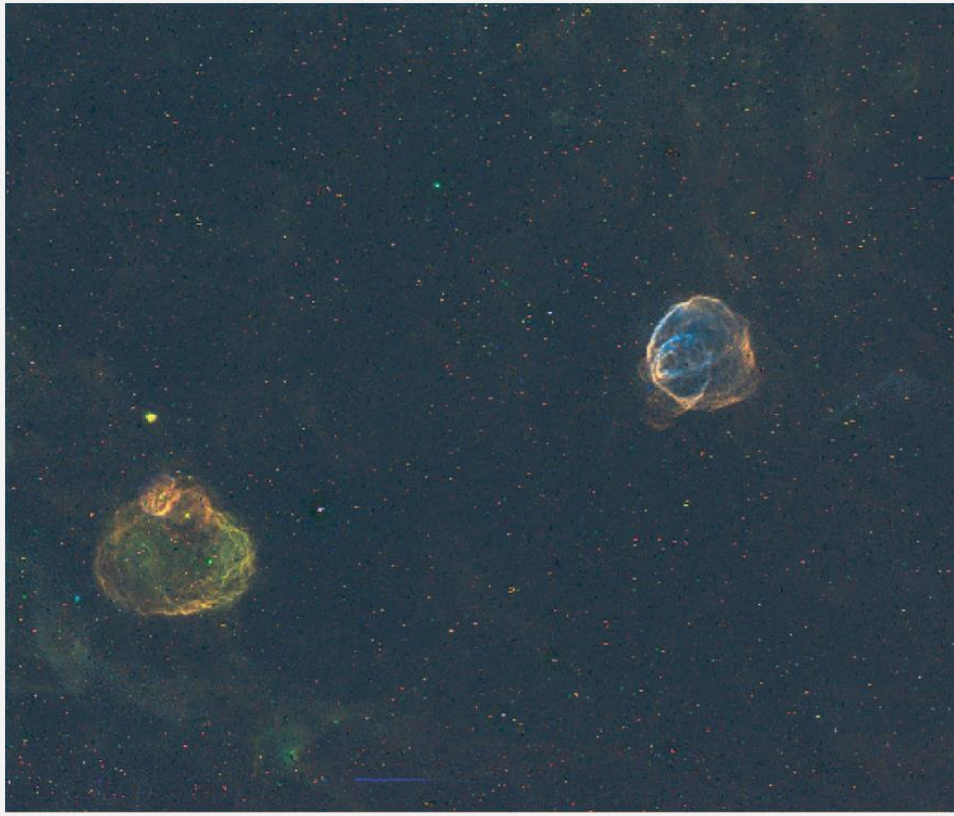
<u>Motivation of this work:</u>

- Develop a SN FB model **considering both kinetic & thermal FBs**.
- First, model SN FB based on **pc-scale simulations**.
  - For **kinetic** feedback model, we run simulations of SNR as shown next.
  - For **thermal** feedback model, we use results of previous works on small-box simulations.
- Second, implement the results into a galactic-scale simulation.

# Superbubble

- When multiple SNe are clustered in space and time, **Superbubbles** are formed.
- Superbubbles are **ubiquitous** because massive ($> 8\ M_{\odot}$) stars are formed in stellar clusters.
- Superbubble's typical size is **100-300 pc.**

- We study dependency of superbubble's momentum on **density**, **metallicity** and **time interval** of SN explosions



Superbubbles in the Large Magellanic Cloud.
Left: DEM L50, Right: DEM L25
(Oey+ 2002)

# Superbubble Simulation: Setup

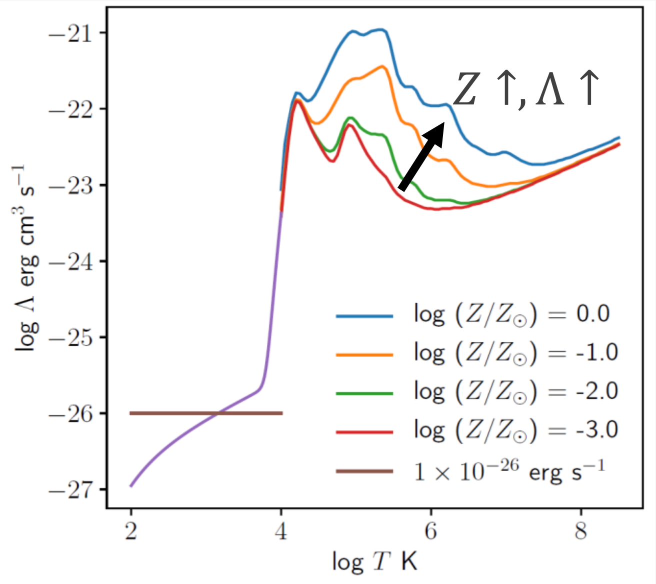
- 3D hydrodynamic simulation using Athena++ (Stone+ 2020)
- Implemented physics:
  - **Turbulence** (Kolmogorov 1941; Larson 1981)
  - **Thermal conduction** (Parker 1953; Spitzer & Härm 1953; Balbus & McKee 1982; El-Badly+ 2019)
  - **Heating** (Kim & Ostriker 2015) constant rate, $1 \times 10^{-26}$ erg s$^{-1}$
  - **Cooling** (Sutherland & Dopita 1993; Koyama & Inutsuka 2002)
- Initial conditions
  - $3 \times 4 \times 3 = 36$ runs

| | |
|---|---|
| $n_H[\mathrm{cm}^{-3}]$ | 0.1, 1, 10 |
| $\log(Z/Z_\odot)$ | -3, -2, -1, 0 |
| $\Delta t_{\mathrm{SN}}[\mathrm{Myr}]$ | 0.01, 0.1, 1 |

| $n_H$ [cm$^{-3}$] | $\Delta x$ [pc] |
|---|---|
| 0.1 | 6 |
| 1 | 3 |
| 10 | 0.75 |

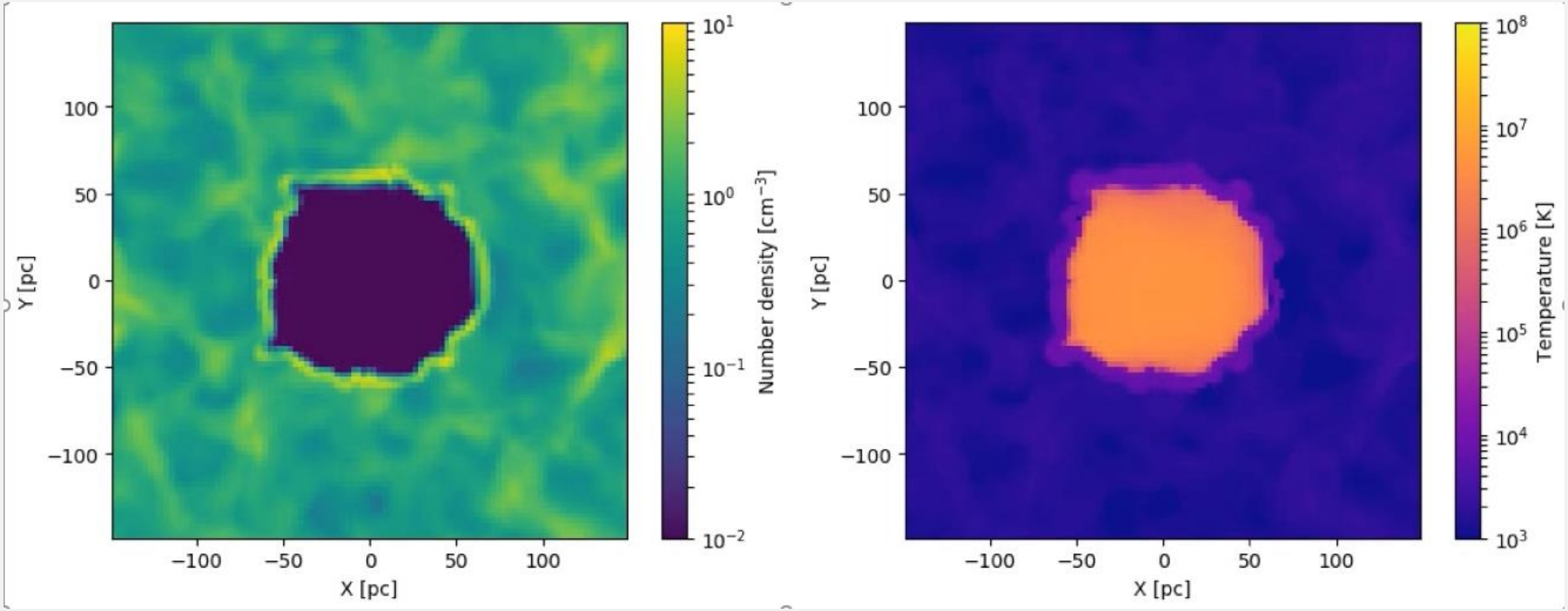
- Thermal energy $E_{SN} = 10^{51}$ erg is injected **10 times** with time interval $\Delta t_{\mathrm{SN}}$.
- In this work, we study **metallicity** dependence using **3D** simulation. (cf. Keller+ 2014; Kim+ 2016; Gentry+ 2019; El-Badly+ 2019)

Cooling function



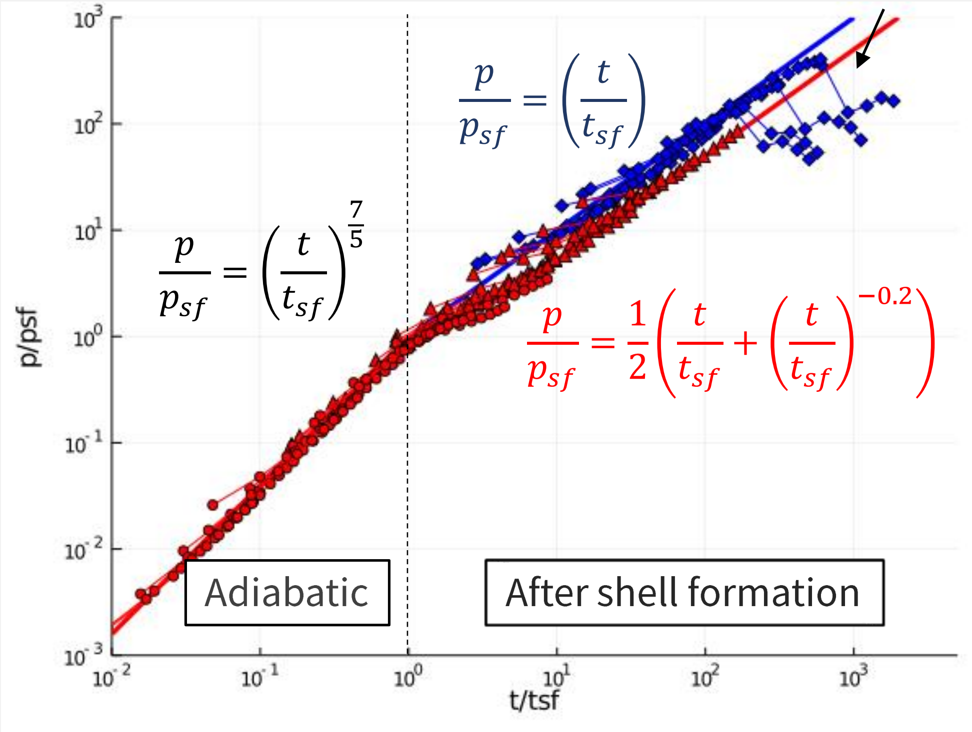
# Superbubble Simulation: Movie

- Movie of $n_H = 1\ \mathrm{cm}^{-3}, Z = Z_{\odot}, \Delta t_{SN} = 0.1\ \mathrm{Myr}$ until $t = 10\ \Delta t_{SN}$.
- SNe occur at the center of the simulation box.
- Superbubble **sweeps up ISM.**
- Almost spherical shell is formed, and it stalls.



# Superbubble Simulation: Result

- **A unified time evolution model of superbubble's momentum** is obtained when the quantities are normalized by values at shell formation time,.
- Momentum per single SN ~ $\mathbf{10^5}\ \boldsymbol{M_\odot}\,\mathbf{km}\ \boldsymbol{s^{-1}}$ (Kim & Ostriker 2015, 2017; Martizzi+ 2015; Fielding+ 2017).




Red: $\Delta t_{\rm SN} < \frac{1}{10} t_{\rm PDS}$ (continuous),

Blue: $\Delta t_{\rm SN} > \frac{1}{10} t_{\rm PDS}$ (intermittent).

$t_{\rm PDS}$: Duration time of Pressure-Driven Snowplow phase

$$p_{sf} = 3.8 \times 10^3 \left(\frac{E_{SN}}{10^{51}\mathrm{erg}}\right)^{1.2} \left(\frac{\Delta t_{SN}}{1\ \mathrm{Myr}}\right)^{-1.2} \left(\frac{n}{1\ \mathrm{cm}^{-3}}\right)^{-0.80} \left(\frac{\Lambda(10^6\mathrm{K})}{10^{-22}\ \mathrm{erg\ cm^3\ s^{-1}}}\right)^{-0.59} [M_\odot \mathrm{km\ s^{-1}}]$$

$$t_{sf} = 1.7 \times 10^4 \left(\frac{E_{SN}}{10^{51}\mathrm{erg}}\right)^{0.28} \left(\frac{\Delta t_{SN}}{1\mathrm{Myr}}\right)^{-0.28} \left(\frac{n_0}{1\ \mathrm{cm}^{-3}}\right)^{-0.71} \left(\frac{\Lambda(10^6 K)}{10^{-22}\ \mathrm{erg\ cm^3\ s^{-1}}}\right)^{-0.42} [\mathrm{yr}]$$

$$\Lambda(T = 10^6 K) = \left(1.876 - 0.827\frac{Z}{Z_\odot}\right)\frac{Z}{Z_\odot} + 10^{-1.33}\ [10^{-22}\ \mathrm{erg\ cm^3\ s^{-1}}]$$

Expanded eqns of Kim & Ostriker '15 adding $\Lambda$ term.

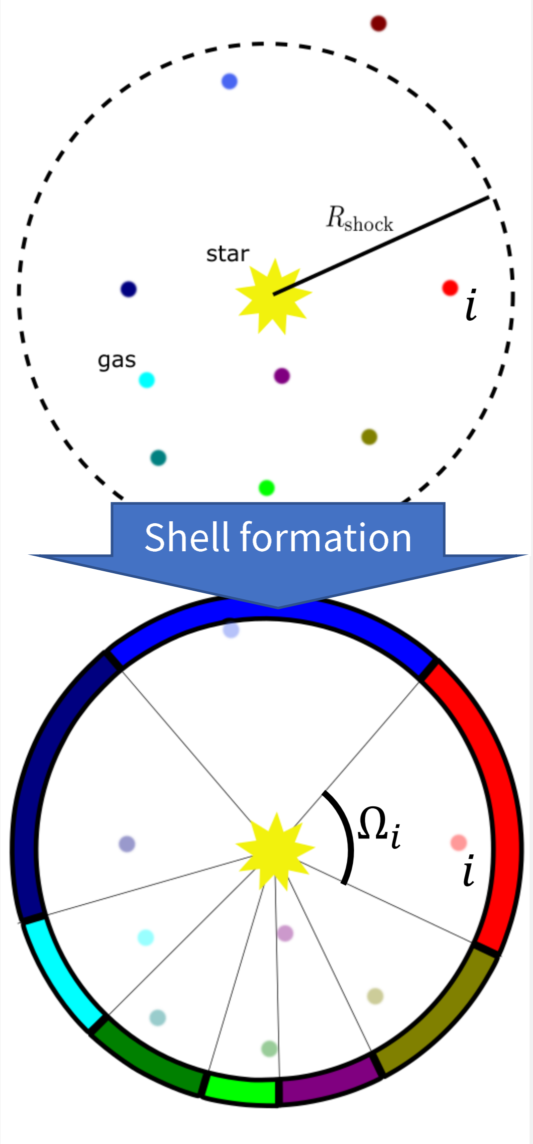
**Obtained the scaling between $p/p_{sf}$ and $t/t_{sf}$.**

# Implementation of Kinetic FB in SPH

- Use the fitting function obtained in the previous slide to estimate the **terminal momentum** of a shell.
- **How to assign the momentum?**

- Use Voronoi tessellation:
  1. Estimate **shock radius and terminal momentum from $\rho, Z, E_{SN}$**.
  2. Get coordinates of neighbor gas particles.
  3. **Voronoi tessellation** on a sphere using STRIPACK algorithm (Renka 1997).
  4. Calculate viewing angle $\Omega$ from the star (SN site).
  5. Assign **momentum,** <u>**mass** and **metals**</u> proportional to the viewing angle.

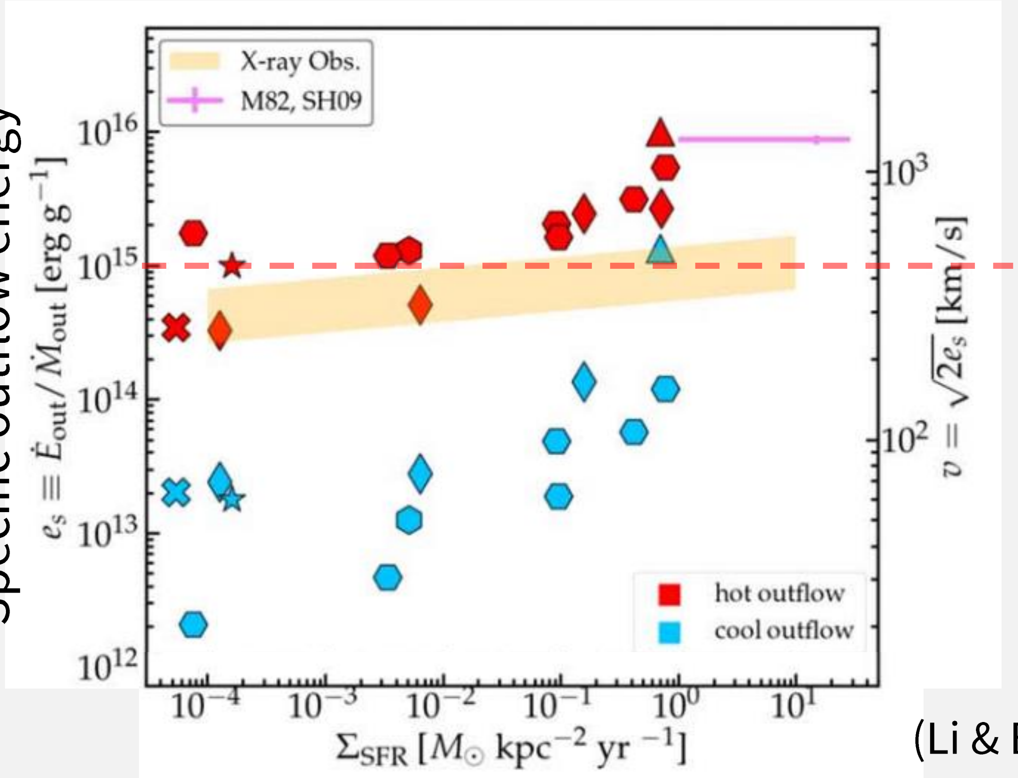
(obtained from CELib (Saitoh 2017))



# Result from small-box simulations

- **How to assign thermal energy?**
- When **cooling mass** is unresolved, thermal feedback is **ineffective (overcooling problem)**.

Specific outflow energy




$T \sim 5 \times 10^6$ K

$$\left( T = \frac{\mu m_p}{k_B} e_s \right)$$

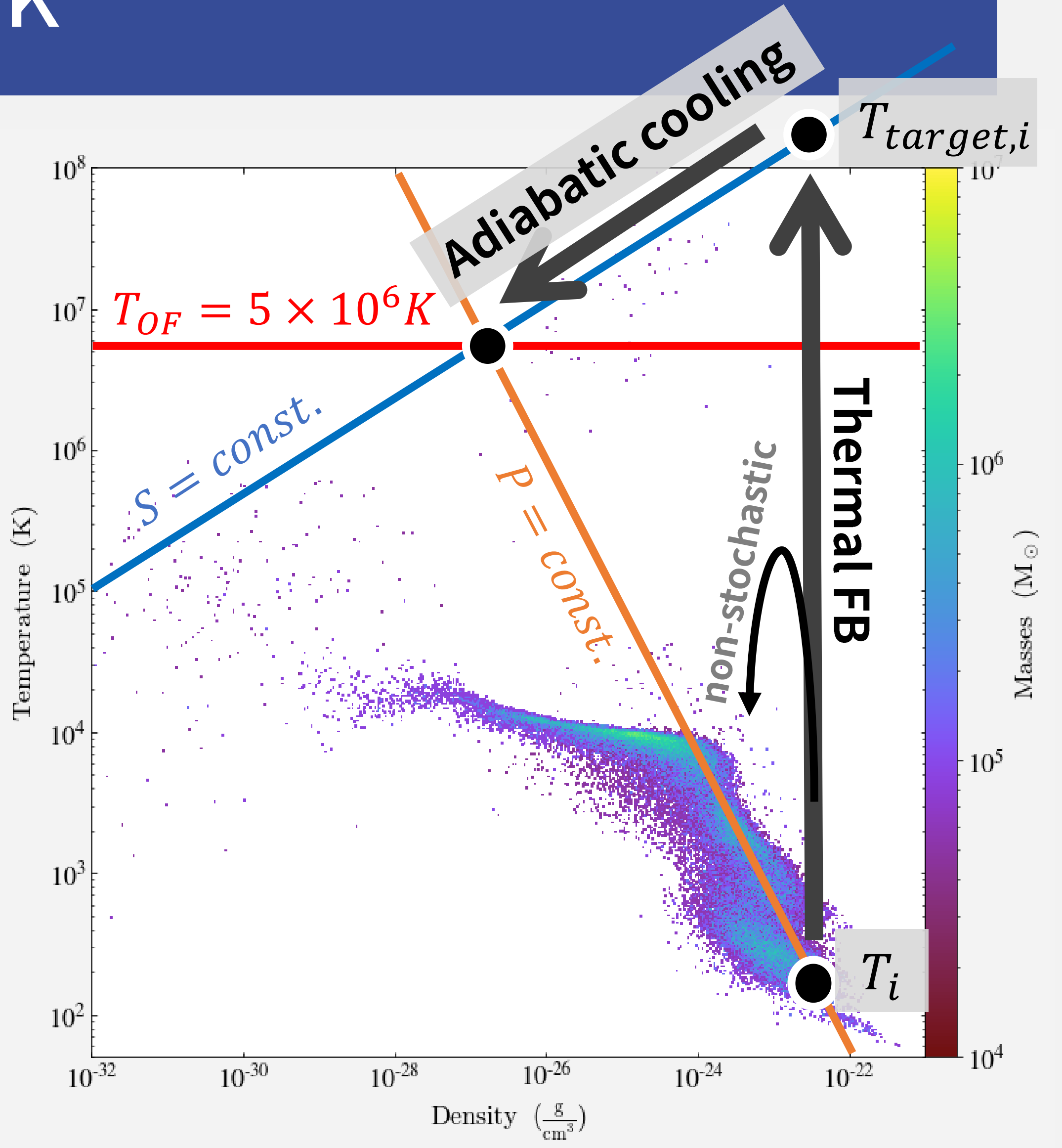
(Li & Bryan 2019)

Star formation surface density

High-resolution small-box simulations show **temperature of hot outflow is typically $5 \times 10^6$ $K$.**

# Stochastic Thermal Feedback

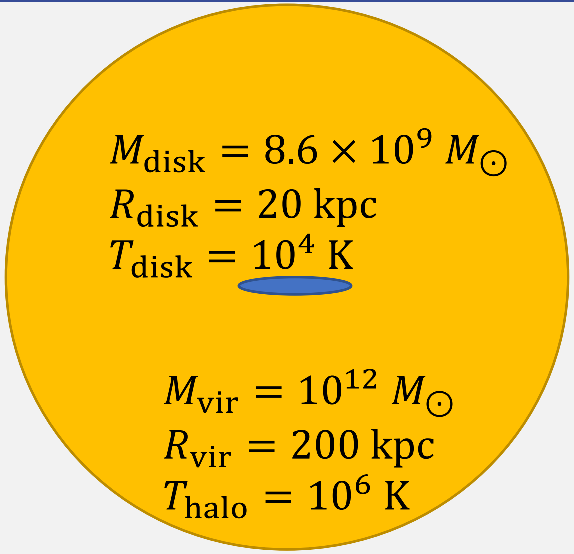
- Updated **stochastic thermal feedback model** (Dalla Vecchia & Schaye 2012).
- Consider an effect of **adiabatic cooling** to reproduce hot outflow with $T_{OF} \sim 5 \times 10^6 K$ ,
- Assume pressure equilibrium between outflow and ISM, the **target temperature is** $T_{\mathrm{target},i} = T_{\mathrm{OF}}^{5/3}\, T_i^{-2/3}$.
- Inject **thermal energy** $\Delta E_i = \frac{m_i}{\mu m_p} k_B T_{\mathrm{target},i}$ with **probability** $P_i = \frac{\Omega_i}{4\pi}\frac{E_{th}}{\Delta E_i}$
- Total thermal feedback energy is $0.7 E_{SN}$.
- Multiphase-ISM model is not used.
- **Cooling is always on**.




**Heats gas up to high temperature by stochastic treatment**

# Isolated Galaxy Test: Setup

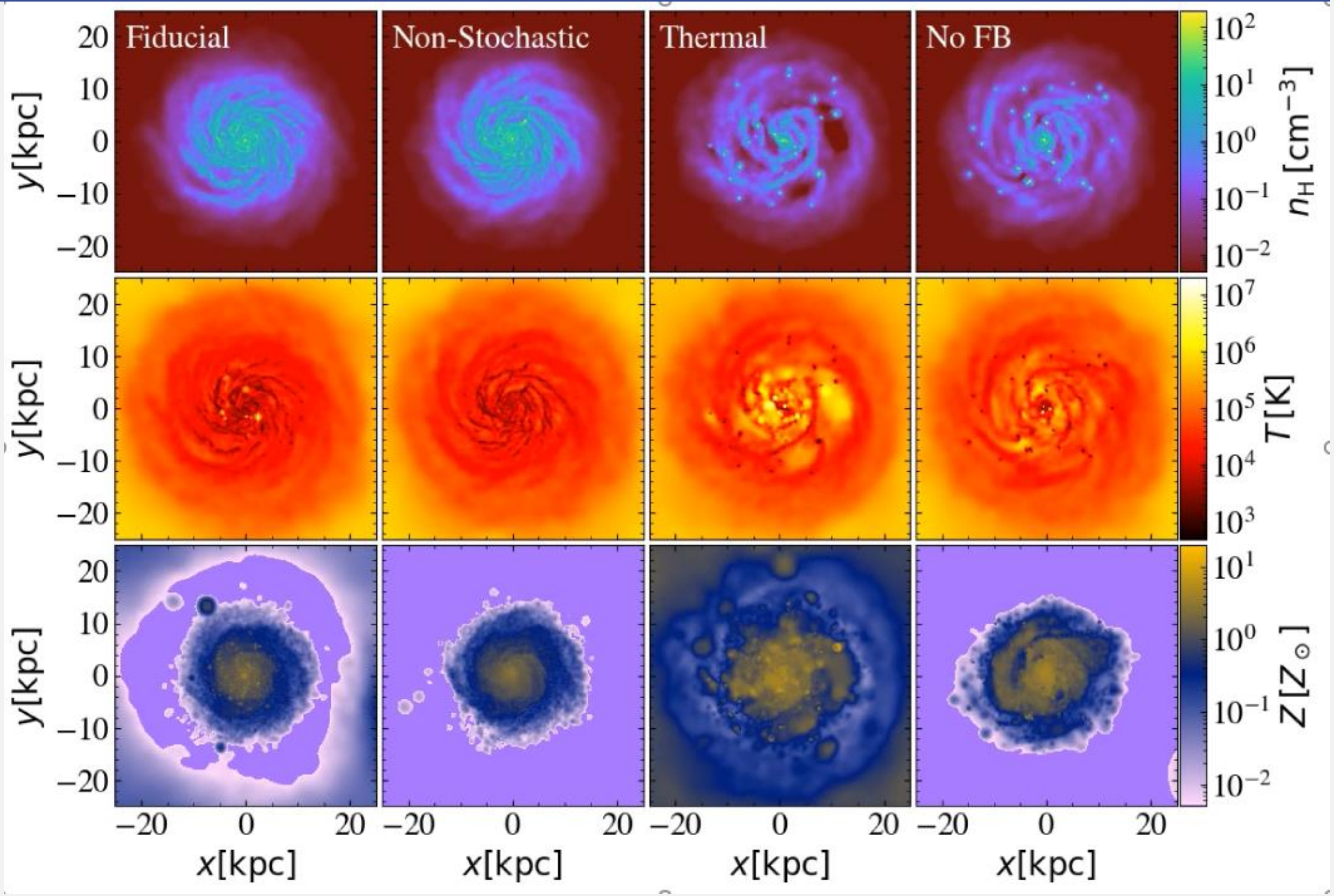
- Isolated galaxy simulation using GADGET3-Osaka (Shimizu+ 2019).
- Initial condition: AGORA disk (Kim+ 2014) + gas halo (Shin+ 2021)
  - $m_{gas} = 8.6 \times 10^4\ M_\odot, m_{DM} = 1.3 \times 10^7\ M_\odot$
- Evolved to 1 Gyr.



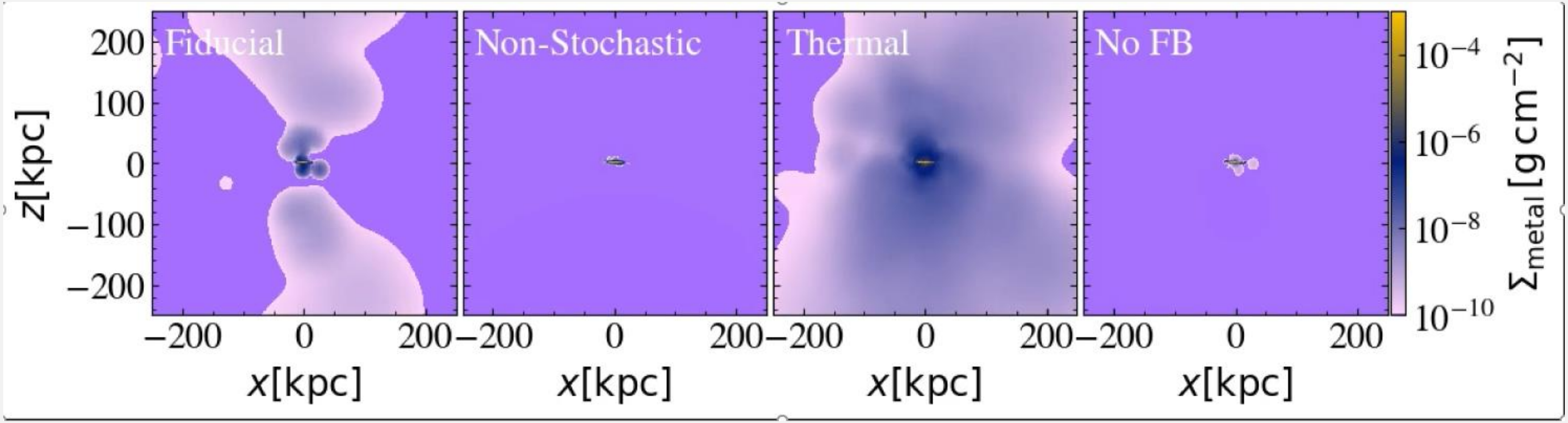

| Name | Description | Purpose |
|---|---|---|
| Fiducial | Fiducial (momentum FB + stochastic thermal FB) | |
| Non-stochastic | Thermal energy is distributed without stochastic treatment | To evaluate the **stochastic treatment** |
| Thermal | Only stochastic thermal FB | To evaluate the **effect of momentum kick** |
| No FB | No feedback | Reference |

# Isolated Galaxy Test: Face-on Movie

- **Without kinetic FB,** gas can easily **collapse** and the disk becomes **clumpy**. (Thermal & No FB)
- **Hot bubbles** are created by stochastic treatment (Fiducial & Thermal)
- Hot gas in No FB run is due to projection effect.

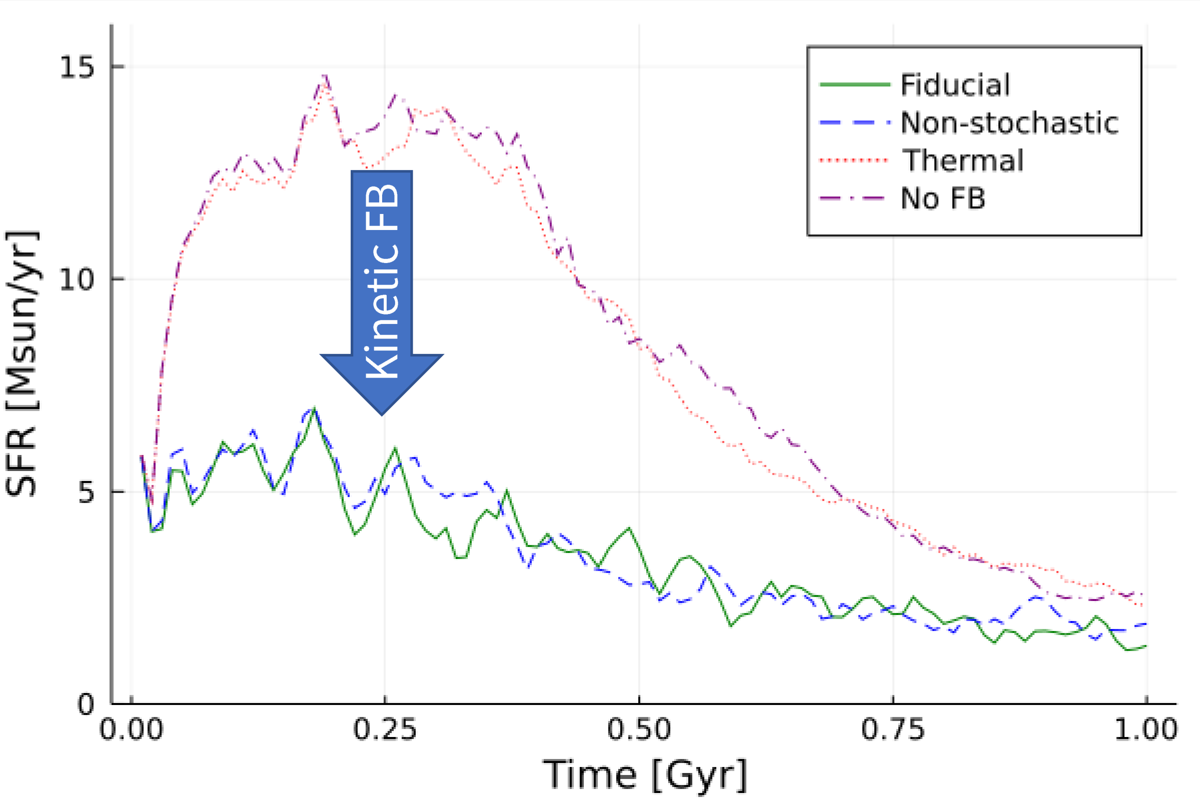


# Isolated Galaxy Test: Edge-on Metal Outflow




- Outflow by stochastic thermal feedback **transports metals up to $R_{vir}$** $(= 200\ \mathrm{kpc})$.
- Gases kicked by kinetic feedback **fall back** to the galaxy.

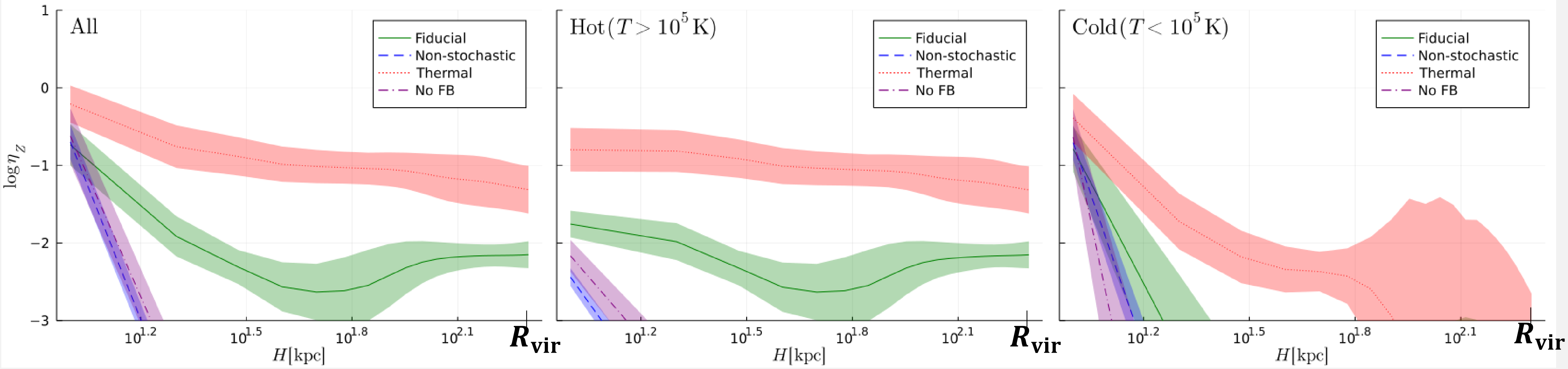
# Isolated Galaxy Test: SFR




- Kinetic FB **suppresses** star formation (Fiducial vs. Thermal vs. No FB).
- Stochastic thermal feedback have **little effect** on star formation (Fiducial vs. Non-stochastic).

# Isolated Galaxy Test: Outflow

Metal loading factor $\left(\eta_Z = \frac{\dot{M}_{Z,out}}{\dot{M}_{Z,SN}}\right)$ profile averaged over $t = 0.5 - 1$ Gyr




- **Stochastic thermal FB** produces **hot ($T > 10^5$ K) outflow** reaches $R_{vir}$.
- Metals are transported by **hot outflow**.

# Summary

- We developed a SN FB model based on pc-scale high-reso simulations.

  ⇨ Obtained $p - t$ scaling ⇨ Implemented in galactic-scale simulation
- Use Voronoi tessellation to model momentum input by superbubble.
- For thermal FB, we updated stochastic FB model to reproduce hot ($T = 5 \times 10^6$ K) outflow.

From isolated-galaxy test,

- Kinetic FB suppresses star formation.
- Stochastic thermal FB produce hot outflow and transport metals up to $R_{\rm vir}$.

- We will use this model in cosmological simulations.